MODULE 1 · LECTURE 10

ICT Applications in Agriculture: Role of IT in Farming and Rural Development

What information and communication technology offers farmers at each stage from planning to selling, how it changes extension and rural services, what the evidence shows, and why it does not always work.

Learning objectives

After this lecture you should be able to:

1. Define ICT and e-agriculture and describe the components of an ICT system.
2. Explain why farming is information-intensive and list farmers' information needs.
3. Compare ICT tools, from radio to smartphones and sensors.
4. Describe the role of IT at each stage of the agricultural value chain and in extension.
5. Explain the role of IT in rural development with Indian examples.
6. Summarise the evidence from well-known case studies and studies of impact.
7. Discuss the benefits and challenges of ICT in agriculture and how to make it work.

Scope: Lectures 1–8 introduced computers, databases, the Internet and programming. This lecture gives an overview of how they are used in farming and rural life. Specific tools are covered later: computer models (Lecture 11), controllers (12), water and nutrient tools (13), smartphone apps (14), geospatial technology (15), decision support and expert systems (16–17), big data and IoT (23), and national digital-agriculture programmes (30–31).

1. ICT and e-agriculture

Information technology (IT) is the use of computers and software to capture, store, process and present information. **Information and communication technology (ICT)** adds the means of **communicating** it — telephone and mobile networks, the Internet, radio, television and satellites. In agriculture the two terms are often used together.

FAO and ITU describe **e-agriculture** as improving agricultural and rural development through better information and communication processes. They use ICT as an umbrella term for all devices, networks, services and applications, from Internet-era technologies and sensors to older aids such as telephones, radio, television and satellites (Food and Agriculture Organization of the United Nations & International Telecommunication Union [FAO & ITU], 2016).

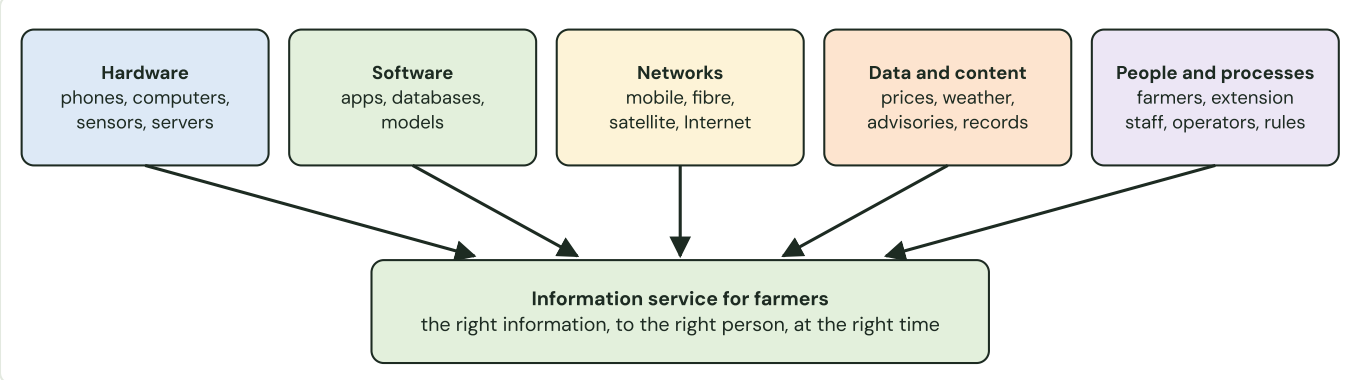


Figure 1. The five components of an ICT system serving farmers. Original diagram.

All five components must work together. A good app is of little use without network coverage, reliable data or trained people to explain it — a lesson repeated in many projects (World Bank, 2017).

2. Why farmers need information

Farming is a chain of decisions made under uncertainty: what to grow, when to sow, how much fertiliser to apply, when to spray, when and where to sell. Each depends on information about weather, soil, pests, prices and policies — information that is often scattered, late or unreliable in rural areas. ICT lowers the cost of getting such information and of acting on it.

Decision	Information needed
What to grow	Suitable crops and varieties, expected prices, water availability
Inputs	Seed and fertiliser rates, dealer locations, prices, subsidies
Crop care	Weather forecasts, pest and disease alerts, irrigation needs
Finance and risk	Credit, crop insurance, government schemes and their status
Selling	Prices in different markets, buyers, storage options, quality standards
Land and rights	Land records, ownership and tenancy documents

3. ICT tools used in agriculture

Tool	Strengths	Limitations	Typical use
Radio and television	Very wide reach; no literacy needed; cheap for the listener	One-way; general, not personal; fixed broadcast times	Farm programmes, weather bulletins
Telephone call centres	Two-way; local language; expert answers	Limited by staff and call volume	Kisan Call Centre, toll-free 1800-180-1551 (Lecture 7)
SMS and voice messages	Reach basic phones; cheap; timely	Short; text needs literacy; risk of being ignored as spam	Weather and pest alerts, scheme updates
Internet kiosks and service centres	Shared access with a trained operator	Need power, connectivity and a viable business	e-Choupal, Common Service Centres
Smartphones and apps	Personal, location-specific, images and video	Cost, skills, data charges	Advisories, prices, record keeping (Lecture 14)
Video	Shows a practice being done; persuasive	Production effort; needs screening or a smartphone	Farmer-to-farmer videos
Social media and messaging groups	Fast, free, peer learning	Misinformation; hard to verify	FPO and village groups
GIS, remote sensing and drones	Field- and region-level maps	Specialised skills and data	Crop monitoring, insurance (Lecture 15)
Sensors and IoT	Continuous automatic data; can trigger control	Cost, maintenance, connectivity	Soil moisture, weather stations (Lectures 12, 23)

4. Role of IT along the agricultural value chain

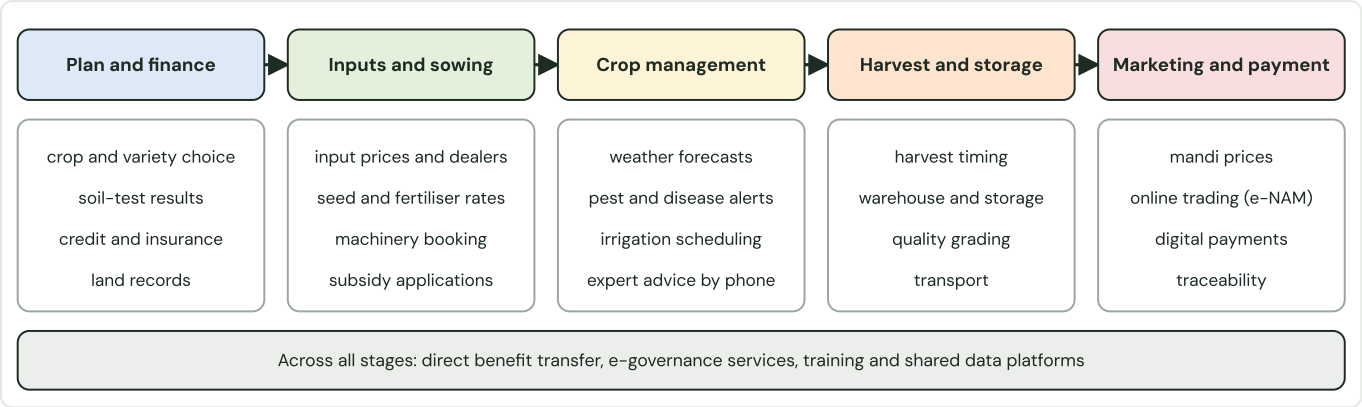


Figure 2. ICT applications at each stage of the agricultural value chain. Original diagram.

- **Planning and finance** — choosing crops and varieties, soil-test-based recommendations (Soil Health Card, Lecture 6), loans, insurance and land records.
- **Inputs and sowing** — finding inputs and their prices, recommended rates, booking machinery from custom-hiring centres, applying for subsidies online.
- **Crop management** — weather and agromet advisories, pest and disease alerts, irrigation scheduling (Lecture 13), expert advice by phone or app.
- **Harvest and storage** — deciding when to harvest, finding warehouse space, grading quality, arranging transport.
- **Marketing and payment** — comparing mandi prices, selling online through e-NAM (Lecture 7), receiving digital payments, and tracing produce from farm to consumer.

Worked example — the value of a timely weather message (illustrative values)
A farmer plans to spray 4 acres; chemical and labour cost ₹1,500 per acre. An SMS forecast warns of heavy rain the next day, which would wash the spray off.
Postponing the spray avoids a loss of 4 × 1,500 = ₹6,000.
If the forecast service costs the farmer nothing, or a few rupees a month, one avoided loss in a season repays it many times over. This is why the benefits of information services usually exceed their delivery cost — provided the forecast is accurate and the farmer trusts it.

5. ICT in agricultural extension

Agricultural extension carries knowledge from research to farmers and brings farmers' problems back to research. Traditional extension relies on field staff visiting villages, and each worker can reach only a limited number of farmers. ICT lets the same knowledge reach many more farmers, faster, and lets farmers ask questions — **e-extension**.

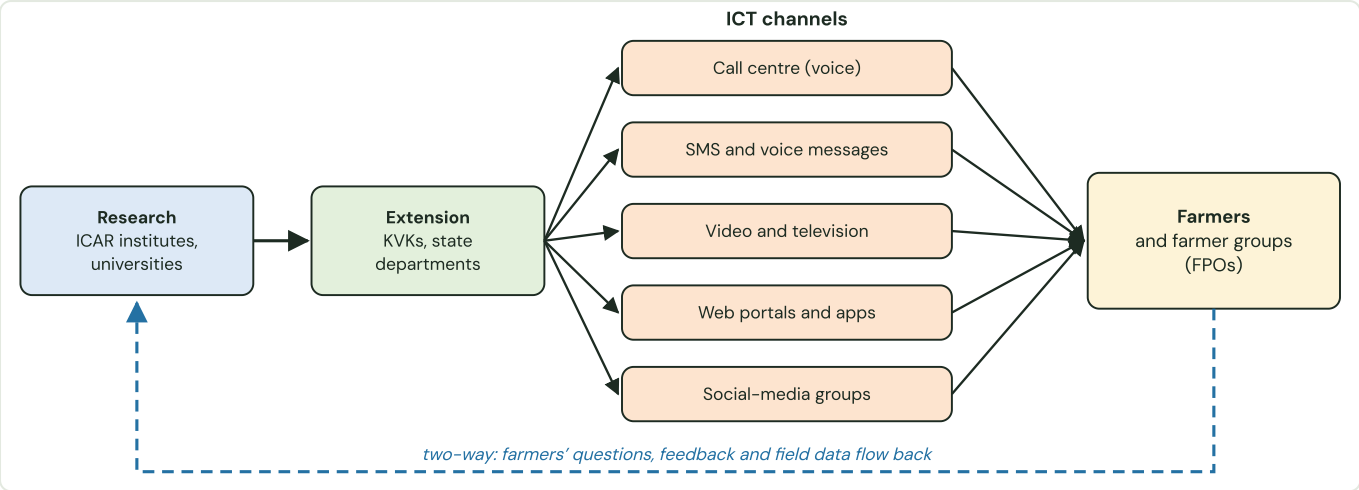


Figure 3. Information flow in ICT-supported extension. Original diagram.

Model	How it works	Example
Call centre	Farmers phone a toll-free number and are answered in their own language; difficult questions go to experts	Kisan Call Centre, started 21 January 2004 (Department of Agriculture and Farmers Welfare, n.d.)
Push messages	Advisories sent as SMS or recorded voice to registered farmers	Weather, pest and scheme messages
Participatory video	Local farmers appear in short videos of good practices, screened in villages by a trained mediator who leads discussion	Digital Green (Gandhi et al., 2007)
Kiosks and infomediaries	A trained local operator gives access to information and services	e-Choupal, Common Service Centres
Apps and portals	Farmers look up advice, prices and services themselves	Lecture 14

6. Role of IT in rural development

Rural development covers incomes, services, rights and quality of life — not only farming. IT supports it by bringing government and business services to the village, making payments direct and transparent, and opening access to learning and health care.

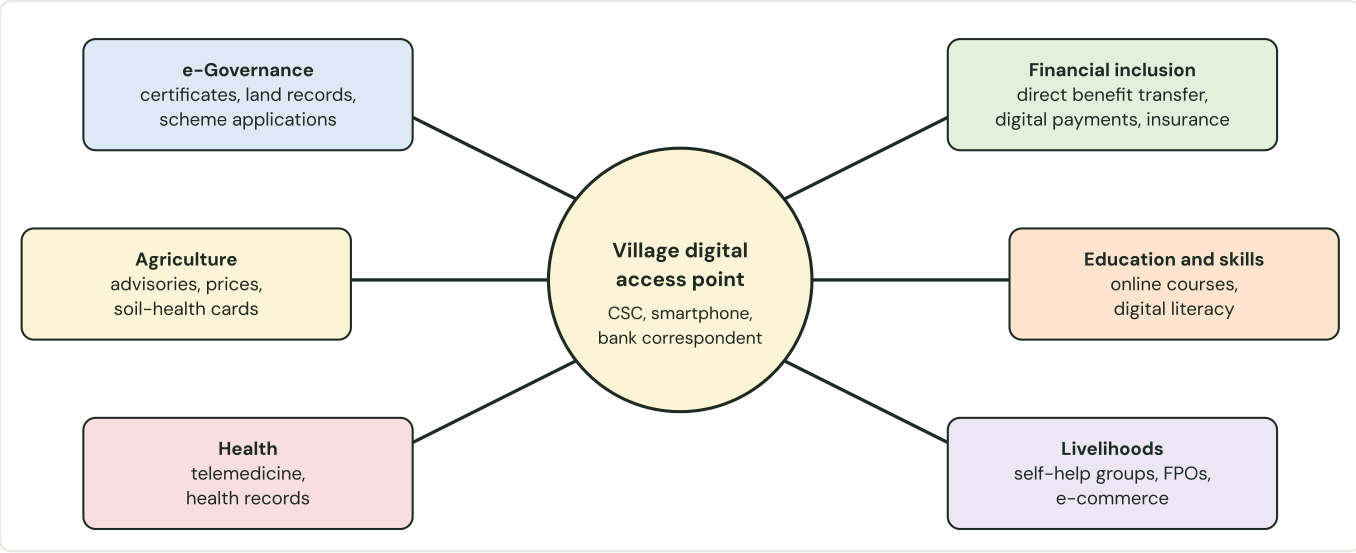


Figure 4. Services delivered through a village digital access point. Original diagram.

Area	Role of IT	Indian example
Service delivery (e-governance)	Certificates, applications and payments at a village centre instead of distant offices	**Common Service Centres (CSCs)**: approved in September 2006 under the National e-Governance Plan and run by local village-level entrepreneurs; 4,67,491 were functional in September 2026, of them 3,68,621 in rural areas (Ministry of Electronics and Information Technology [MeitY], n.d.)
Direct benefit transfer (DBT)	Money paid straight into beneficiaries' bank accounts, reducing delays and leakage	**PM-KISAN**: launched on 24 February 2019, it pays eligible farmer families ₹6,000 a year in three instalments. Over ₹3.70 lakh crore reached more than 11 crore farmer families through the first 20 instalments, and the 21st, in November 2025, transferred ₹18,000 crore to 9 crore farmers (Press Information Bureau [PIB], 2025)
Land records	Computerised, printable records that farmers can obtain quickly	**Bhoomi**, Karnataka (case study below)
Financial inclusion	Digital payments, banking through local correspondents, online insurance claims	UPI payments; crop-insurance portals
Education and skills	Online courses and digital-literacy training	University and KVK online courses
Health	Teleconsultation from village centres and phones	Telemedicine services
Livelihoods	Market access for self-help groups and FPOs; online selling	e-commerce and e-NAM (Lecture 7)

National Digital India and digital-agriculture programmes, including AgriStack, are covered in detail in Lectures 30 and 31.

7. Case studies

7.1 Mobile phones and Kerala's fishing markets

Mobile phone service spread along the Kerala coast between 1997 and 2001. Jensen (2007) used survey data from fishing markets and found that once fishermen and wholesalers adopted mobile phones, the differences in fish prices between nearby markets fell sharply, the waste of unsold fish was eliminated, and prices followed the **law of one price** almost exactly. Both fishermen and consumers gained. The lesson: when information is costly, markets work poorly; cheap communication lets sellers go where the buyers are.

7.2 ITC e-Choupal

ITC Limited set up Internet kiosks in villages, operated by a local farmer, giving price, weather and farming information and an alternative channel for selling produce directly to ITC. ITC reports about 6,100 kiosks covering over 35,000 villages in 10 states and serving over 4 million farmers (ITC Limited, n.d.). In Madhya Pradesh, where the kiosks began for soybean in October 2000, Goyal (2010) found a significant rise in the soybean price farmers received after the kiosks were introduced, and an increase in the area sown to soybean — evidence that better price information and a competing buyer improved rural markets.

7.3 Bhoomi — computerised land records in Karnataka

Farmers need the **Record of Rights, Tenancy and Crops (RTC)** for bank loans and many other purposes. Under the manual system it was kept by village accountants; getting a copy took 3 to 30 days and often a bribe. The Bhoomi project computerised about 20 million land records of 6.7 million farmers, and printed RTCs became available at kiosks in all 177 taluk offices for a fee of ₹15, with a second screen so that the farmer could watch the transaction (Chawla, n.d.). Changes of ownership (mutations) are handled first-come, first-served and can be tracked, reducing the scope for bribery. The lesson: IT can make government records open and services predictable.

7.4 Digital Green — participatory video

Digital Green, developed with the GREEN Foundation in Karnataka, records short videos of local farmers demonstrating good practices and screens them in villages with a TV and DVD player, led by a locally hired mediator. In a four-month trial in 16 villages (1,070 households), it increased the adoption of certain practices six to seven times compared with classical person-only extension; the authors called the results preliminary but promising (Gandhi et al., 2007). The lesson: technology works best when it uses trusted local people and social learning.

7.5 What the wider evidence shows

Reviewing studies from sub-Saharan Africa and India, Fabregas, Kremer and Schilbach (2019) report that digital agricultural information increased yields by **4%** and the odds of adopting recommended inputs by **22%**. These effects are modest in absolute terms but large compared with the very low cost of sending messages; the authors estimate that benefits likely exceed costs by an order of magnitude. They also note reasons for caution: some farmers ignore messages from unknown senders, illiterate farmers struggle with text and voice menus, messages can be confusing, and information alone cannot overcome missing credit, inputs or infrastructure.

8. Benefits and challenges

Benefits	Challenges
Timely, location-specific information on weather, pests and prices	Weak connectivity and power supply in remote villages
Better prices and less dependence on intermediaries	Cost of devices and data for small farmers
Extension reaches far more farmers at low cost per farmer	Low digital literacy; content not in local languages
Transparent, faster government services and payments	Misinformation, fraud and spam reduce trust
Records and data for credit, insurance and planning	Projects that stop when pilot funding ends
Two-way communication: farmers' problems reach researchers	Data privacy; many apps and portals that do not share data

The **digital divide** — the gap in access to and use of ICT — remains wide. At the end of June 2026 India had 49.07 Internet subscriptions per 100 people in rural areas against 129.03 in urban areas (Telecom Regulatory Authority of India [TRAI], 2026). Gaps also exist by income, age, gender and education, so services designed only for smartphone users can leave many farmers behind.

Making ICT work for farmers: use local languages, voice and video; keep messages short, specific and from a trusted source; combine digital tools with extension staff and local mediators; collect feedback and test what works; let systems share data; plan for running costs beyond the pilot; and protect farmers' personal data (Lecture 6).

9. Key takeaways

- ICT combines computing with communication; e-agriculture uses it to improve agricultural and rural development through better information.
- An ICT service needs hardware, software, networks, good content and trained people working together.
- Farming decisions — what to grow, how to manage the crop, when and where to sell — depend on information, and ICT lowers its cost.
- ICT supports every stage of the value chain and makes extension two-way and far-reaching.
- In rural development, IT delivers services through Common Service Centres, direct benefit transfers such as PM-KISAN, and computerised land records.
- Evidence from Kerala fisheries, e-Choupal, Bhoomi and Digital Green, and the meta-analysis by Fabregas et al. (2019), shows real but context-dependent benefits.
- Connectivity, cost, skills, language, trust and sustainability determine whether ICT reaches the farmers who need it most.

10. Examination questions

Short-answer questions (2 marks)

Q1. Define ICT.

Technologies for capturing, storing, processing and communicating information — computers, software, mobile and Internet networks, radio, television and satellites.

Q2. What is e-agriculture?

The use of ICT to improve agricultural and rural development through better information and communication processes (FAO and ITU).

Q3. List the components of an ICT system.

Hardware, software, networks, data and content, and people and processes.

Q4. What is e-extension?

Agricultural extension delivered or supported through ICT channels such as call centres, SMS, video, apps and portals.

Q5. What is e-Choupal?

ITC Limited's network of village Internet kiosks giving farmers price, weather and farming information and a direct selling channel.

Q6. What is the RTC, and which project computerised it in Karnataka?

The Record of Rights, Tenancy and Crops, a land record; the Bhoomi project.

Q7. What are Common Service Centres?

Village-level ICT access points, run by local entrepreneurs, that deliver government, financial and business services.

Q8. What is direct benefit transfer? Give an agricultural example.

Payment of benefits directly into beneficiaries' bank accounts; PM-KISAN.

Q9. What did Jensen (2007) find about mobile phones in Kerala's fisheries?

Price differences between markets fell sharply, waste was eliminated and both fishermen and consumers gained.

Q10. What is the digital divide?

The gap in access to and use of ICT between groups, such as rural and urban people.

Long-answer questions (5–10 marks) — answer outlines

Q1. Define ICT and e-agriculture. Explain the components of an ICT system with a diagram.

Definitions (Section 1); Figure 1; role of each component; why all must work together.

Q2. Describe the ICT tools used in agriculture, with their strengths and limitations.

Table in Section 3: radio/TV, call centres, SMS/voice, kiosks, smartphones, video, social media, GIS, sensors.

Q3. Explain the role of IT at different stages of the agricultural value chain.

Figure 2; planning, inputs, crop management, harvest, marketing; worked example on the value of information.

Q4. Discuss the role of ICT in agricultural extension, with Indian examples.

Figure 3; e-extension models — Kisan Call Centre, SMS advisories, Digital Green, e-Choupal, apps; two-way flow.

Q5. Explain the role of IT in rural development.

Figure 4; CSCs, DBT (PM-KISAN), Bhoomi land records, financial inclusion, education, health, livelihoods.

Q6. Describe two ICT case studies in agriculture and discuss the challenges of using ICT in Indian agriculture.

Any two of Kerala fisheries, e-Choupal, Bhoomi, Digital Green with outcomes; evidence from Fabregas et al.; benefits vs challenges table; digital divide.

11. Glossary

Term	Meaning
CSC	Common Service Centre — a village-level digital service point
DBT	Direct benefit transfer — payment straight into a beneficiary's bank account
Digital divide	Gap in access to and use of ICT between groups
e-Agriculture	Use of ICT to improve agricultural and rural development
e-Extension	Extension supported by ICT channels
e-Governance	Use of ICT to deliver government services and information
Extension	Transfer of knowledge between research and farmers
ICT	Information and communication technology
Infomediary	A person or centre that helps others access and use information
IVR	Interactive voice response — automated phone menus and recorded messages
Kiosk	A shared computer or Internet access point, often run by an operator
Law of one price	In a well-informed market, the same good sells for the same price in different places, apart from transport cost
Mutation	Recording a change in land ownership in the land records
Participatory video	Videos made with and featuring local people
RTC	Record of Rights, Tenancy and Crops — a land record in Karnataka
Value chain	The stages through which a product passes from inputs to the consumer
VLE	Village-level entrepreneur who runs a Common Service Centre

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All diagrams are original. Case studies are summarised in original language from the cited sources. Values in the worked example are illustrative.

Practice MCQs

Q1. ICT differs from IT mainly by adding

(a) data storage
(b) communication networks
(c) programming
(d) printing

Q2. Which is NOT a component of an ICT system?

(a) Hardware
(b) Networks
(c) Data and content
(d) Crop rotation

Q3. According to Jensen (2007), mobile phones in Kerala's fishing markets led to

(a) higher waste of fish
(b) larger price differences between markets
(c) smaller price differences and less waste
(d) no change

Q4. ITC e-Choupal is

(a) a crop-insurance scheme
(b) a network of village Internet kiosks run by ITC
(c) a soil-testing laboratory
(d) a government call centre

Q5. The Bhoomi project in Karnataka computerised

(a) soil-health cards
(b) land records (RTC)
(c) mandi prices
(d) weather data

Q6. Digital Green spreads farming practices mainly through

(a) satellite images
(b) participatory videos shown by local mediators
(c) radio broadcasts
(d) printed leaflets

Q7. Common Service Centres are

(a) agricultural research stations
(b) village-level digital service points
(c) cold-storage units
(d) seed banks

Q8. PM-KISAN pays eligible farmer families

(a) ₹2,000 a year
(b) ₹6,000 a year in three instalments
(c) ₹12,000 a year
(d) ₹6,000 once

Q9. In the meta-analysis by Fabregas et al. (2019), digital agricultural information increased yields by about

(a) 4%
(b) 22%
(c) 40%
(d) 100%

Q10. Which channel best suits farmers who cannot read text messages?

(a) E-mail
(b) Voice calls and video
(c) Printed reports
(d) Spreadsheets

Q11. The gap in access to ICT between rural and urban people is called the

(a) digital divide
(b) value chain
(c) law of one price
(d) information asymmetry index

Q12. Moving government services and payments online is called

(a) e-commerce
(b) e-governance
(c) e-learning
(d) e-mail

Answers

Q1 — (b) communication networks
Q2 — (d) Crop rotation
Q3 — (c) smaller price differences and less waste
Q4 — (b) a network of village Internet kiosks run by ITC
Q5 — (b) land records (RTC)
Q6 — (b) participatory videos shown by local mediators
Q7 — (b) village-level digital service points
Q8 — (b) ₹6,000 a year in three instalments
Q9 — (a) 4%
Q10 — (b) Voice calls and video
Q11 — (a) digital divide
Q12 — (b) e-governance